

Revision of Pattern with while loops →

```

4      *
3      * *
2      * * *
1      * * * *
0      * * * * *
  
```

(Space)

+2 +2 +2 +2
(1, 3, 5, 7, 9)

Star	space
1	4
3	3
5	2
7	1

St	sp
9	0

```

int n = 5;
int row = 1, star = 1, space = n - 1;
  
```

```

while (row <= n) { ← row
  
```

```

    int i = 1;
    while (i <= space) {
      cout << " ";
      i++;
    }
  
```

// stars

```

    int j = 1;
    while (j <= star) {
      cout << "* ";
      j++;
    }
  
```

```

    star = star + 2;
    space = space - 1;
    row = row + 1;
    cout << endl;
  
```

```

} ← row
  
```

H.W.

0 space + 9 stars

1 space + 7 stars

2 space + 5 stars

3 space + 3 stars

4 space + 1 star

```

* * * * * * * * *
 * * * * * * *
  * * * * * *
   * * * * *
    * * * *
     * * *
      * *
       *
  
```

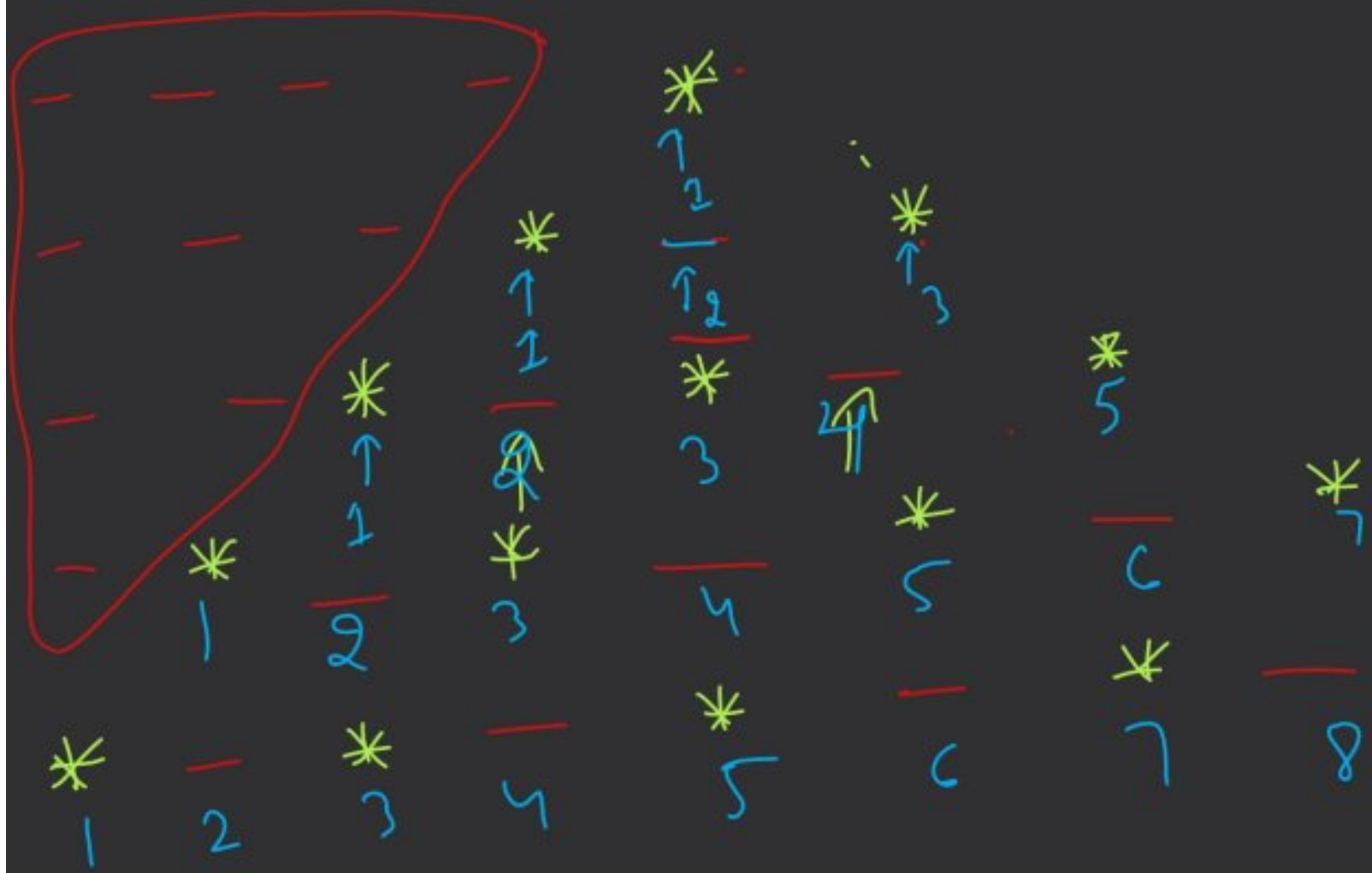
int row = 1, star = 2 * n - 1

int space = 0;

star = star - 2;

space = space + 1;

row = row + 1;



(Blue $\rightarrow$ j values)

```

int row = 1, star = 1, space = 4;
while (row <= n) {
    int i = 1;
    while (i <= space) {
        cout << " ";
        i++;
    }
    int j = 2;
    while (j <= star) {
        if (j % 2 == 0) {
            cout << " ";
        } else {
            cout << "x";
        }
        j++;
    }
    cout << "\n";
    row++;
    star = star + 2;
    space = space - 1;
}

```


Ans.



space = -1

star = 5

12 K

$n=5$, $space = 2 * n - 3$

Row = 1, star = -1

while (row <= n) {

int i = 1;

// stars while (i <= star) {

cout << " * ";

i++;

// space int j = 1;

while (j <= space) {

cout << " ";

j++;

if (row == n) {

k = 2;

// star int k = 1;

while (k <= star) {

cout << " * ";

k++;

cout << endl;

row++;

space = space - 2;

star = star + 1;

~~✱~~ ✱ ✱
 ✱ ✱ ✱
 ✱ ✱ ✱

(1 —————> 10)

✱
 ✱ ✱
 ✱ ✱ ✱
 ✱ ✱ ✱ ✱

* * * *

* * * *
↑ ↑ ↑ ↑

1 2 3 4 5
* * * * *

int i = 1;
while (i <= star) {
 cout << " * ";
 i++;
}

int k = 1; ←
if (row == n) { k = 2; }
while (k <= star) {
 cout << " * ";
 k++;
}

23
456

Ques.

+ 2
+ 2
+ 2

5

4

3

2

1



(|| Try to think
of the initial
conditions)

n=5

1 <= -1 X

-1, 1, 3, 5, 7

H.W. Do a dry
run for
n=7


```
int n = 5, stars = 5, space = -1
```

```
while (row <= n) {
```

```
    // star int i = 1;  
    while (i <= stars) {
```

```
        cout << " * ";
```

```
        i++;
```

```
    }
```

```
    // space int j = 1;  
    while (j <= space) {
```

```
        cout << " ";
```

```
        j++;
```

```
    }
```

```
    // star
```

```
    int k = 1;
```

```
    if (row == 1) k = 2;
```

```
    while (k <= stars) {
```

```
        cout << " * ";
```

```
        k++;
```

Star ↓ ← -1

Space ↑ ← +2

Row ↑ ← +1

Solid
Rhombus

4 space + 5 star
3 spa + 5 star
2 spa + 5 star
1 space + 5 star
0 space + 5 star

```

  _ _ _ _ * * * *
 _ _ _ * * * *
_ _ * * * *
_ * * * *
* * * *
  
```

```

while (row <= n) {
    int i = 1;
    while (i <= space) {
        cout << " ";
        i++;
    }
}
  
```

```

    int j = 1;
    while (j <= star) {
        cout << "* ";
        j++;
    }
  
```

```

int n = 5, row = 1,
star = n, space = n - 1;

cout << endl;
row++;
space--;
}
  
```


⑦

DancingStar.java > DancingStar > main(String[])

```
1 public class DancingStar {
    Run | Debug
2     public static void main(String[] args) {
3         int n = 5;
4         int star = 1;
5         int space = n - 1;
6         int row = 1;
7
8         while(row <= n) {
9             int i = 1;
10            while(i <= space) {
```

(Do a dry Run
with n=8)
(row=1 → 8)

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Code + ▾ □ ✕ ... | [] ✕

```
PS C:\Users\91950\Desktop\coding_blocks\Lecture_7> cd "c:\Users\91950\Desktop\coding_blocks\Lecture_7\" ; if ($?) { java  
vac DancingStar.java } ; if ($?) { java DancingStar }
```

```
*
* *
* * *
* * * *
* * * * *
```

→ Output

DancingStar.java > DancingStar > main(String[])

```
1 public class DancingStar {  
    Run | Debug  
2     public static void main(String[] args) {  
3         int n = 5;  
4         int star = 1;  
5         int space = n - 1;  
6         int row = 1;  
7  
8         while(row <= n){  
9             int i = 1;  
10            while(i <= space){
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Code + - [] [X] ... [] [X]

```
PS C:\Users\91950\Desktop\coding_blocks\Lecture_7> cd "c:\Users\91950\Desktop\coding_blocks\Lecture_7\" ; if ($?) { java  
vac DancingStar.java } ; if ($?) { java DancingStar }
```

```
*  
* *  
* * *  
* * * *  
* * * * *  
* * * * *
```

→ Output